

Trading Strategies — Living Document

A running catalogue of every signal/strategy we've considered, built, tested, or shelved — with **status and evidence**, so we refine over time instead of re-litigating. Update the changelog at the bottom on every material change.

Golden rule: nothing earns risk until it survives **point-in-time features + friction-aware, regime-split backtesting**. The broad-sentiment edge looked great and died exactly this way — see `backtest/engine.py`.

Status legend

	Meaning
✔ Validated	Survived friction-aware, regime-split backtest (still preliminary — see caveats)
● Marginal	Real effect, but fragile net of costs / not sign-stable
✗ Failed	Tested and does not work (kept here so we don't retry it)
🧪 Untested	Built and wired, awaiting data to evaluate
💡 Idea	Backlog — not yet built
🔄 In progress	Data/build task underway

Progress at a glance

Last updated: 2026-06-01. Update this section whenever a status, metric, or next-step changes.

Strategy scoreboard

Strategy	Class	Status	Headline result	Next action
<code>retail_fear</code> (capitulation / bear-extreme / pessimism-fade)	Macro overlay	✔ Validated	+0.2–0.47pp/10d excess vs buy-&-hold ; AUDJPY best (win 88%); VIX gate ~4×	live via MT5 <code>RedditMacro_EA</code> ; longer history; vol-target sizing
<code>rrai_momentum_long</code> , <code>froth_high_long</code>	Macro overlay	✗ No edge	positive net but ~0 excess vs buy-&-hold (just bull drift)	dropped
<code>rrai_euphoria_short</code> + all macro shorts	Macro overlay	✗ Failed	–1.5 to –3pp excess (shorting the bull)	dropped
<code>dump_fade_DELETION_short</code>	Equity	🧪 Untested	awaiting coverage (<code>gone_frac>0</code> in 267 rows)	re-backtest at ~30–40% author coverage; express via puts
<code>fade_organic_short</code>	Equity	● Marginal	+0.83%/5d but P3 flips	retry via puts + VIX/vol gate
<code>pump_long</code> (ride)	Equity	✗ Failed	+3% but regime-flips, t=0.75	— drop
<code>dump_fade_short</code> (naive)	Equity	✗ Failed	–9.9% (squeezed)	superseded by deletion-gated
<code>organic_long</code>	Equity	✗ Failed	–4.2%, sign-stable loser	inverse → <code>fade_organic_short</code>

Build & data progress

Component	Status	Detail
Feature store (<code>feature_daily</code> / <code>aggregate_daily</code>)	✅ Done	~77k ticker-day rows, point-in-time
Friction-aware backtester (<code>backtest/engine.py</code>)	✅ Done	costs + regime split + no-lookahead
<code>/signals</code> feed (multi-strategy, json mt5)	✅ Done	strategy-tagged; live, currently flat
Macro strategy R&D (<code>backtest/macro_research.py</code>)	✅ Done	only retail_fear beats buy-&-hold; momentum/froth no edge
MT5 framework (<code>SignalClient.mqh</code> + <code>RedditMacro_EA.mq5</code>)	✅ Done	strategy-selectable thin client; demo-only
Author profiling coverage	📊 8%	3,823 / 47,625; 184 gone (116 del / 68 susp); target ~40%
Deletion-gated short testability	📊 Surfacing	<code>gone_frac>0</code> in 267 feature rows; too sparse to trade yet
Idea backlog	💡 7 ideas	squeeze · options-flow · hype-cycle · sentiment-extremes · rotation · novelty · coordination
Equity/options paper client (Alpaca)	💡 Planned	not started

Methodology (non-negotiables)

1. **No lookahead** — features at date D use only data $\leq D$; entries at the first close *after* D.
2. **Frictions** — round-trip spread (small-caps wide), short borrow, halts, capacity.
3. **Regime split** — net return per sub-period; sign-stability is the bar, not a big average.
4. **Shorts via PUTS**, not naked stock — penny shorts have squeeze/borrow/halt tails.
5. **Overlapping-window caveat** — daily signals with N-day forward returns inflate t-stats.
6. **Benchmark macro vs buy-and-hold** — in a bull market "long anything" looks like alpha; a macro long must beat *unconditional drift* (excess), not zero. (Cost us 5 false "survivors".)

Asset-class reality (what this data can/can't trade)

- **Small/micro-cap & meme equities** — primary signal (retail flow moves price).
- **Single-name options** — primary *expression* (convex longs, defined-risk shorts).
- **Index / risk-FX** — *macro overlay only* (retail positioning as a contrarian factor).
- **FX majors / "Reddit sentiment on SPY as direction"** — no edge; don't build it.

1. Macro overlay — Retail Risk-Appetite Index (RRAI)

Aggregate retail positioning as a **contrarian** factor at extremes. Built posts-only + ratios (coverage-stationary); 90-day trailing percentile. `analytics/features.py::aggregate_daily`.

Macro R&D verdict (`backtest/macro_research.py` , **excess over buy-and-hold**). A bull market made *every* macro long look like a winner and *every* short a loser — until benchmarked against unconditional drift. Net of that, only the **contrarian retail-fear** family beat buy-and-hold. RRAI-momentum and froth had ~0 excess (pure drift); all shorts were -1.5 to -3pp (fighting the trend). **One real macro edge, three formulations of it.**

✓ retail_fear — buy risk when retail capitulates (served live)

- **Thesis:** retail max-bearish (RRAI pct low / bear-share extreme) + genuine fear (VIX high) → contrarian bounce. Captured 3 ways (capitulation, bear-extreme, pessimism-fade) — all positive-excess, mutually corroborating.
- **Spec (shipped):** long when `rrai_pct ≤ 0.15` **AND** `VIX ≥ 18`; hold ~10d; size by extremity; **fear-side only**.
- **Evidence (12mo, net, 10d):** excess over buy-&-hold SPY +0.17pp / QQQ +0.23pp / **AUDJPY +0.47pp** (win 88%); `bear_extreme` +0.24/+0.32/+0.36pp corroborates. **VIX gate ~4x'd it** (SPY +1.56%/10d in fear vs +0.42% calm). Modest **timing tilt**, not standalone alpha (buy-&-hold already won 66–75%).
- **Expression:** FX **AUDJPY** (best leg), index CFDs US500/USTEC. Served via `/signals` (strategy tag `retail_fear`); traded by `mql5/RedditMacro_EA.mq5` (demo).
- **Caveats:** single ~12mo bull regime, $n \approx 43\text{--}66$ /bucket, overlapping windows; edge is small in absolute terms.
- **Next:** leverage-language enrichment (calls/puts, margin/YOLO); longer history (data-limited); vol-target sizing; combine capitulation+bear-extreme into one composite trigger.

✗ Macro candidates that failed (kept so we don't retry)

- `rrai_euphoria_short`, `bull_extreme_short`, `froth_high_short`, all momentum-shorts: **–1.5 to –3pp excess** — shorting a bull market.
- `rrai_up_momentum_long`, `froth_high_long`: **~0 excess** — just drift, no signal.

2. Equity cross-sectional (per ticker × day)

From `analytics/features.py::feature_daily` (velocity, accel, breadth, concentration/HHI, sentiment trajectory, author young/gone/suspended mix, novelty).

✗ pump_long — ride concentrated promotion

- **Thesis:** concentrated mention spike (few authors, high volume) pops short-term.
- **Result:** net +3%/5d but **sign-flips** (P1 +15.8 → P3 –4.6), win 36%, $t=0.75$. The crude "+6.45%" was a no-lookahead/no-cost/single-regime artifact. **Not robust**.

✗ dump_fade_short — short a decelerating pump (naive)

- **Result:** net **–9.9%** (P1 –29%). Shorting pumps on deceleration gets squeezed/run over. Wrong trigger; brutal without defined risk.

✗ organic_long — buy broad attention spikes

- **Result:** sign-stable **–4.2%** net ($t=-4.25$, $n=321$). Buying broad/news-driven attention tops is a *consistent loser* → the inverse is the candidate (below).

● fade_organic_short — fade broad attention spikes

- **Thesis:** broad attention spikes mean-revert down (inverse of the robust `organic_long` loss).
- **Result:** net +0.83%, but **flips in P3** (win 46%) — the reversion is real but shorting costs + recent weakening eat it. **Marginal**. Revisit via **puts** (cheaper than borrow) and with a VIX/vol gate.

🧪 dump_fade_DELETION_short — short a pump whose promoters vanished (THE thesis)

- **Thesis:** concentrated pump + promoters now **deleted/suspended** → manufactured → fade the collapse. This is the original A2 idea, and the *reason* we track account lifecycle.
- **Status:** built & wired; **awaiting author coverage**. As of 2026-06-01: 8% of 47k authors profiled, 184 gone accounts found (~4.7% base rate), `gone_frac > 0` in 267 feature rows — surfacing but too sparse to intersect the pump filter yet. Background `author_age_backfill` sweep grinding; scheduler rebuilds features each cycle.

- **Next:** re-run `python main.py --backtest` as coverage climbs (~30–40%); consider prioritising profiling of authors on pump-candidate tickers to make it testable sooner. Express via puts.

3. Idea backlog (💡 not yet built)

- **Squeeze setup (long):** Reddit squeeze chatter (Shortsqueeze/SqueezePlays) + **short-interest / float / borrow-fee** → squeezable. *Needs external SI/borrow data.*
- **Options-flow signal:** parse contract-level chatter (unusual_whales/options) → retail-gamma/positioning proxy. *Needs contract parsing.*
- **Hype-cycle lifecycle:** model attention as nascent→acceleration→peak→fade; long the *organic* acceleration, fade the *concentrated* peak. Breadth-vs-concentration is the discriminator.
- **Conditioned sentiment extremes (single-name):** linear sentiment is dead, but *extreme concentrated euphoria at a price high* (short) / *capitulation after a sharp drop* (bounce) may be non-linear edges.
- **Cross-sectional attention rotation:** rank meme universe by abnormal velocity z-score; long top-decile / short bottom-decile, beta-hedged with SPY; baskets from co-mention clusters.
- **Novelty / first-mention alpha:** the "discovery" moment (silent → sudden organic burst).
- **Coordination graph (short):** rigorous version of the deletion short — cluster density + temporal bursts + duplicate text + new-account% + gone%.

4. Data gaps blocking strategies

Gap	Unblocks
Intraday prices (have EOD only)	timing pump/dump entries/exits (they move intraday)
Short interest / float / borrow fee	squeeze setup; short feasibility
Options chains / IV	options expression; options-flow signal
Longer history (data-limited to ~12mo)	true multi-regime validation of the RRAI overlay
Author coverage (8% → target ~40%+)	the deletion-gated short (<i>in progress</i>)

5. Execution stack

- `/signals feed` (`analytics/signals.py` , `GET /signals json|mt5`) — live, serves the capitulation overlay. ✅
- **MT5 EA** (`mq15/RedditRRAI_EA.mq5`) — thin client for the index/FX overlay; demo-only. ✅
- **Alpaca / IBKR paper client** — for the equity/options legs. 💡 planned.

Changelog

- **2026-06-01** — Macro R&D + MT5 framework. Benchmarked candidates vs buy-and-hold: only the `retail_fear` contrarian family beats it (best AUDJPY +0.47pp/10d excess); momentum/froth = no edge, all shorts fail. Built `SignalClient.mqh` + `RedditMacro_EA.mq5` (strategy-selectable), multi-strategy `/signals feed`; retired `RedditRRAI_EA.mq5`.
- **2026-06-01** — Initial document. Phase 0 (feature store + friction-aware backtester) shipped. RRAI capitulation-long validated (best via AUDJPY, VIX-gated); pump-ride/naive-dump-short/organic-long failed; fade-organic marginal; deletion-gated short awaiting author coverage (8%). `/signals` + MT5 EA live.